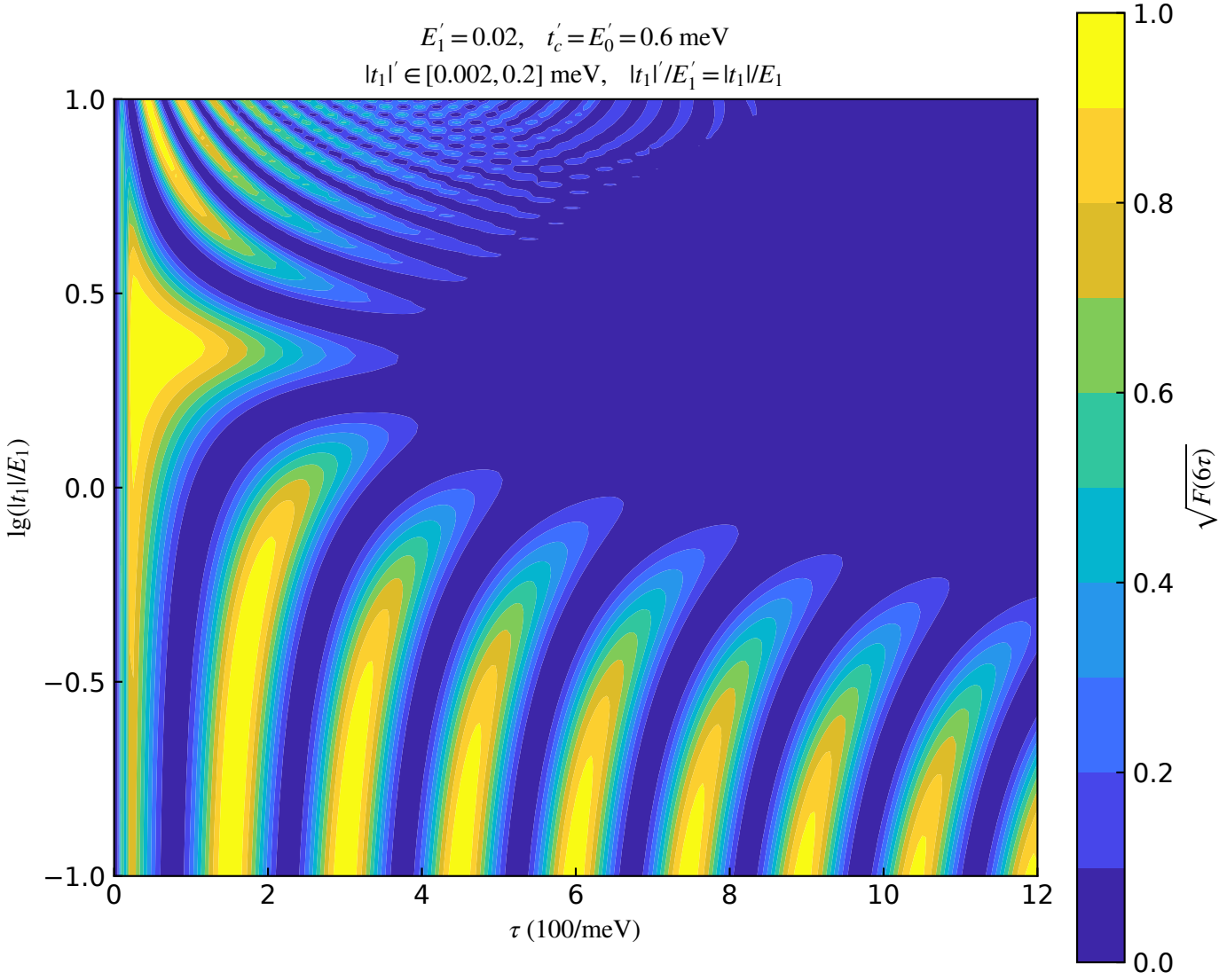
$$\sqrt{F}: \quad K = 2A(\mathbf{p}) = A(2\mathbf{p})$$



Matrix amplitudes absorb the factor 2; the six-stage protocol and duration are unchanged.